

Graph Neural Networks for Neuroimaging-Based Parkinson's Disease Analysis: A Comprehensive Review

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Abstract

Parkinson's Disease (PD) presents significant diagnostic challenges due to the complexity of neuroanatomical changes and the subtle progression of motor and cognitive symptoms. Traditional deep learning approaches, while effective, often fail to capture the complex, non-Euclidean structural and functional relationships inherent in brain connectomes [5]. This review systematically examines the emergence of Graph Neural Networks (GNNs) as a transformative tool for PD analysis using Magnetic Resonance Imaging (MRI). We evaluate state-of-the-art architectures, including Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) [1], focusing on their ability to model brain networks from multi-modal data. By synthesizing findings from recent high-impact studies, we highlight the superior performance of GNNs in differential diagnosis, stage prediction, and identifying biomarkers for deep brain stimulation [8,9]. Furthermore, we discuss the integration of explainable AI (XAI) within graph frameworks to enhance clinical interpretability. This review outlines current limitations and proposes future research directions to accelerate the clinical adoption of graph-based diagnostic systems.

Keywords: Parkinson's Disease; Graph Neural Networks; Neuroimaging; Deep Learning; Brain Connectomes; Multi-modal Fusion; Graph Attention Networks (GATs); Explainable AI (XAI); Medical Image Segmentation; Medical Image Classification.

1. Introduction

Parkinson's Disease (PD) is a neurodegenerative disorder characterized by dopaminergic neuron loss. While MRI is a cornerstone of diagnostic research, extracting biomarkers remains challenging due to scanner bias and data imbalance. Recent shifts in medical AI have moved from standard CNNs toward architectures that better reflect biological structures. Unlike grid-based CNNs, Graph Neural Networks (GNNs) represent the brain as a graph of nodes and edges, capturing complex, long-range dependencies. This review evaluates graph-based PD analysis, emphasizing multi-modal fusion and explainable frameworks for clinical insight.



2. Related Work

2. Methodology and Graph-Based Architectures

The selection of literature for this review focused on high-impact contributions that bridge the gap between geometric deep learning and clinical neuroimaging. Research was prioritized from prestigious venues such as CVPR, ECCV, and MICCAI to ensure technical rigor. The following subsections categorize the core GNN methodologies identified in the literature for Parkinson's Disease analysis.

2.1. Graph Convolutional Networks (GCN)

Graph Convolutional Networks represent the foundational approach for analyzing brain connectivity by generalizing convolutional operations to non-Euclidean data [3]. In PD research, GCNs are primarily utilized to analyze functional connectivity networks derived from rs-fMRI, where nodes represent brain regions and edges represent temporal correlations. These models effectively capture the global topological alterations associated with dopaminergic depletion. Furthermore, relational GCNs have been explored to model the multi-relational data inherent in patient-specific clinical metadata [5].

2.2. Graph Attention Networks (GAT)

While GCNs treat all neighboring nodes with equal importance, Graph Attention Networks (GATs) introduce an attention mechanism that allows the model to learn the relative importance of different brain regions [1]. This is particularly relevant for PD, where certain nodes, such as the substantia nigra and basal ganglia, hold higher diagnostic weight. Recent studies have implemented geometric-weighted GATs [1] to identify functional networks specifically associated with gait impairment, providing a more granular look at motor symptom progression. Additionally, unsupervised GAT clustering has shown promise in segmenting medical images without the need for extensive manual labeling [11,12], addressing the data scarcity challenge in rare parkinsonian syndromes.

2.3. Multi-modal and Heterogeneous Graph Learning

A major advancement in the field is the shift toward omni-modal biomedical representations [2]. Unlike single-modality models, heterogeneous graph learning integrates disparate data—such as structural T1-weighted MRI, diffusion tensor imaging (DTI), and genomic profiles—into a unified structure. For instance, modality-prompted networks create comprehensive patient profiles by coupling structural and functional connectomes [6]. This integration significantly improves differential diagnosis between idiopathic PD and Parkinsonism-Plus syndromes, such as Multiple System Atrophy (MSA) [7], by leveraging adaptive fusion to capture complementary imaging biomarkers.

2.4. Explainability and Graph Pooling

To overcome the "black box" nature of deep learning, explainable graph pooling (XGNN) methods [4] highlight specific brain sub-networks driving diagnostic decisions. By employing contrastive graph pooling, researchers can visualize discriminative patterns of structural atrophy or metabolic changes [7]. These frameworks utilize attention-guided modules to identify relevant features within the graph, ensuring outputs align with established neurological pathways [9]. Such transparency is essential for the clinical validation and ethical deployment of AI-driven systems in neurodegenerative disease management.

3. Results

The performance of the Graph Neural Network (GNN) framework was evaluated across three primary anatomical views—Axial, Coronal, and Sagittal—using the [1] backbone architecture. Unlike the original xGW-GAT implementation [1], which focuses on local functional connectomes with prior annotated regions of interest (ROIs), our approach utilizes whole-image $N \times N$ node embeddings and local feature extraction to achieve high-fidelity classification of Parkinson's Disease (PD).

View	Split	Accuracy (Mean $\pm$ Std)	F1 Score (Mean $\pm$ Std)	Best Accuracy
Axial	0	0.9420 $\pm$ 0.0224	0.9352 $\pm$ 0.0347	0.9584
	1	0.9455 $\pm$ 0.0110	0.9453 $\pm$ 0.0145	0.9576
	2	0.9554 $\pm$ 0.0074	0.9546 $\pm$ 0.0098	0.9635
	3	0.9613 $\pm$ 0.0089	0.9579 $\pm$ 0.0113	0.9705
	4	0.9421 $\pm$ 0.0102	0.9398 $\pm$ 0.0163	0.9486
Coronal	0	0.8553 $\pm$ 0.0260	0.8249 $\pm$ 0.0370	0.8853
	1	0.8623 $\pm$ 0.0426	0.8344 $\pm$ 0.0660	0.9110
	2	0.8340 $\pm$ 0.0135	0.8120 $\pm$ 0.0176	0.8479
	3	0.8595 $\pm$ 0.0231	0.8263 $\pm$ 0.0229	0.8810
	4	0.8473 $\pm$ 0.0309	0.8144 $\pm$ 0.0580	0.8791
Sagittal	0	0.8827 $\pm$ 0.0442	0.8727 $\pm$ 0.0586	0.9247
	1	0.8737 $\pm$ 0.0375	0.8627 $\pm$ 0.0513	0.9152
	2	0.8825 $\pm$ 0.0155	0.8658 $\pm$ 0.0350	0.8992
	3	0.8765 $\pm$ 0.0434	0.8381 $\pm$ 0.0590	0.9146
	4	0.8827 $\pm$ 0.0432	0.8683 $\pm$ 0.0596	0.9312

Summary Global Metrics:

- **Axial:** Mean Best Acc: 0.9597 ($\pm$ 0.0081) | Overall Best Model Mean Acc: 0.9578
- **Coronal:** Mean Best Acc: 0.8809 ($\pm$ 0.0225) | Overall Best Model Mean Acc: 0.8624
- **Sagittal:** Mean Best Acc: 0.9170 ($\pm$ 0.0121) | Overall Best Model Mean Acc: 0.8875

Figure 1 Experimental Results on [1] back bone architecture with baseline and pruned versions of our MRI dataset, for K=4 pruned versions of the baseline dataset, per split.

Experimental data reveals that the Axial view provides the most discriminative features for PD detection, achieving a mean accuracy of 0.9487 ± 0.0084 and a peak best accuracy of 0.9705 (Split 3). While prior research [1] reports accuracy around 91.6% using multimodal MRI, our results suggest that dense graph connections within single-view whole-image embeddings can exceed these benchmarks, particularly in the Axial plane. The Sagittal view followed with a mean best accuracy of 0.9170 ± 0.0121 , while the Coronal view proved the most challenging with a mean best accuracy of 0.8809 ± 0.0225 .

The significant performance gap between the Axial and Coronal views—nearly 7% in mean accuracy—indicates that feature selection and graph pruning are critical for views where anatomical landmarks may be less distinct. These results establish a competitive baseline that demonstrates the efficacy of the GNN backbone in capturing subtle structural signatures of PD across different anatomical orientations.

4. Discussion

GNNs effectively model the non-Euclidean nature of brain connectivity, facilitating the integration of structural and functional data. Our analysis using the [1] backbone shows that the axial view provides the most discriminative $N \times N$ node connections for Parkinson's detection. However, model generalizability across scanners and the "black box" nature of deep layers remain significant hurdles. Future work must prioritize explainable graph pooling (XGNN) to identify localized, clinically relevant biomarkers.

5. Conclusions

GNNs represent a significant advancement in Parkinson's Disease analysis by capturing complex systemic neurodegeneration. While our findings indicate high performance in axial-plane classification, the field's future lies in omni-modal biomedical representation and interpretable attention mechanisms. Expanding these models to include longitudinal data will be vital for predicting disease progression and personalizing treatment strategies, directly supporting global health initiatives.

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